

Internet Protocols: exercise 1

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Exercise 1:

Άσκηση 1:

Σας έχουν δοθεί οι εξής IP διευθύνσεις:

- A. 130.20.14.12 με Subnet Mask 255.255.248.0
- B. 180.1.15.128 με Subnet Mask 255.192.0.0
- C. 230.12.10.32/16
- D. 192.168.1.10/24

Για τις παραπάνω διευθύνσεις απαντήστε στις κάτωθι ερωτήσεις:

1. Ποια είναι η **διεύθυνση δικτύου**;
2. Ποια είναι η **broadcast address**;
3. **Γράψτε την διεύθυνση με CIDR notation** ή γράψτε την **Subnet Mask** (αν είναι σε CIDR notation)
4. Ποιος είναι ο **αριθμός των interfaces** που μπορούν να υποστηρίξουν;

A.

1. -130.20.14.12 in binary: 10000010.00010100.00001110.00001100 -255.255.248.0 in binary: 11111111.11111111.11111000.00000000 - 10000010.00010100.00001110.00001100 AND 11111111.11111111.11111000.00000000 ----- 10000010.00010100.00001000.00000000 Which is 130.20.8.0	2. - OR between network and Wildcard which is: 00000000.00000000.00000111.11111111 10000010.00010100.00001000.00000000 OR 00000000.00000000.00000111.11111111 ----- 10000010.00010100.00001111.11111111 Which is 130.20.15.255
3. CIDR Notation: 130.20.14.12/21	4.Available hosts: $2^{**}(11)-2=2046$ hosts

B.

1. -180.1.15.128 in binary: 10110100.00000001.00001111.10000000 -255.192.0.0 in binary: 11111111.11000000.00000000.00000000 - 10110100.00000001.00001111.10000000 AND 11111111.11000000.00000000.00000000 ----- 10110100.00000000.00000000.00000000 Which is 180.0.0.0	2. - OR between network and Wildcard which is: 00000000.00111111.11111111.11111111 10110100.00000000.00000000.00000000 OR 00000000.00111111.11111111.11111111 ----- 10110100.00111111.11111111.11111111 Which is 180.63.255.255
3. CIDR Notation: 180.1.15.128/10	4. Available Hosts: $2^{**}(22)-2=4.194.302$ hosts

C.

1. -230.12.10.32 in binary: 11100110.00001100.00001010.00100000 -subnet Mask /16 in binary: 11111111.11111111.00000000.00000000 - 11100110.00001100.00001010.00100000 AND 11111111.11111111.00000000.00000000 ----- 11100110.00001100.00000000.00000000 Which is 230.12.0.0	2. - OR between network and Wildcard which is: 00000000.00000000.11111111.11111111 11100110.00001100.00000000.00000000 OR 00000000.00000000.11111111.11111111 ----- 11100110.00001100.11111111.11111111 Which is 230.12.255.255
3. Subnet Mask: 255.255.0.0	4. Available Hosts: $2^{**}(32-16)=2^{**}(16)=65.534$ hosts

D.

1. - 192.168.1.10 in binary: 11000000.10101000.00000001.00001010 - Subnet Mask /24 in binary: 11111111.11111111.11111111.00000000 - 11000000.10101000.00000001.00001010 AND 11111111.11111111.11111111.00000000 ----- 11000000.10101000.00000001.00000000	2. - OR between network and Wildcard which is: 00000000.00000000.00000000.11111111 11000000.10101000.00000001.00000000 OR 00000000.00000000.00000000.11111111 ----- 11000000.10101000.00000001.11111111 Which is 192.168.1.255
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Which is 192.168.1.0	
3. Subnet Mask: 255.255.255.0	4. Available Hosts: $2^{32-24}-2=2^8-2=254$

Exercise 2:

Άσκηση 2: Σας έχει δοθεί το εξής δίκτυο κλάσης B: 156.132.0.0. Σας ζητείται να σπάσετε το δίκτυο αυτό χρησιμοποιώντας **subnetting** (subnet mask), ώστε να το χωρίσετε σε τουλάχιστον 80 δίκτυα, που το καθένα να έχει τουλάχιστον 130 hosts. Ποια **subnet mask** θα χρησιμοποιήσετε; **Αιτιολογήστε την απάντησή σας.**
Σημείωση: Υπάρχει παραπάνω από μία λύσεις. Χρειάζεται να δώσετε μόνο μία.

To solve this problem, we need to divide the given Class B network (156.132.0.0) into at least 80 subnets, each supporting at least 130 hosts. We will determine the appropriate subnet mask to achieve this.

1. Determine the number of bits needed for subnets:
 - We need at least 80 subnets. The formula to find the number of subnets is $2^n \geq \text{number of subnets}$.
 - $2^7=128$, which is greater than 80. So, we need 7 bits for subnetting.
2. Determine the number of bits needed for hosts:
 - Each subnet must support at least 130 hosts. The formula to find the number of hosts is $2^h - 2 \geq \text{number of hosts}$.
 - $2^7 - 2 = 126$, which is less than 130. $2^8 - 2 = 254$, which is greater than 130. So, we need 8 bits for hosts.
3. Calculate the subnet mask:
 - A Class B network has a default subnet mask of 255.255.0.0 (/16).
 - We are borrowing 7 bits for subnetting, so the new subnet mask will be $16+7=23$ bits.
 - The subnet mask in dotted decimal notation is 255.255.254.0 (/23).
4. Verify the solution:
 - With a /23 subnet mask, each subnet can support $2^9 - 2 = 510$ hosts, which is more than enough for 130 hosts.
 - The number of subnets is $2^7=128$, which is more than the required 80.

Therefore, the appropriate subnet mask to use is 255.255.254.0 (/23). This will allow you to create at least 80 subnets, each supporting at least 130 hosts.

Exercise 3:

Άσκηση 3: Όπως είδαμε στο εργαστήριο, τα μηνύματα του πρωτοκόλλου DHCP βρίσκονται μέσα στα options. Χρησιμοποιώντας το RFC 2132 (<https://datatracker.ietf.org/doc/html/rfc2132>) να εξηγήσετε για κάθε μήνυμα στο DHCP.pcap τα διάφορα options. Ανοίξτε το DHCP.pcap χρησιμοποιώντας το wireshark (File→Open). Τι κάνει το καθένα και ποια είναι η σύνταξή του.

<https://datatracker.ietf.org/doc/html/rfc2132>. The first bullet of each option contains where that option is mentioned in RFC 2132

Packet 1:

The image shows a Wireshark packet capture of a DHCP Discover message (Packet 1). The packet list on the left shows four packets: 1. DHCP Discover (0.0.0.0 to 255.255.255.255), 2. DHCP Offer (192.168.0.1 to 192.168.0.10), 3. DHCP Request (0.0.0.0 to 255.255.255.255), and 4. DHCP ACK (192.168.0.1 to 192.168.0.10). The packet details pane on the right shows the structure of the DHCP Discover message. The UDP payload is 272 bytes. The Dynamic Host Configuration Protocol (Discover) section shows the message type as Boot Request (1), hardware type as Ethernet (0x01), and hardware address length as 6. The Transaction ID is 0x00003d1d. The Bootp flags are 0x0000 (Unicast). The Client IP address is 0.0.0.0. The Your (client) IP address is 0.0.0.0. The Next server IP address is 0.0.0.0. The Relay agent IP address is 0.0.0.0. The Client MAC address is Grandstream_01:fc:42 (00:0b:82:01:fc:42). The Client hardware address padding is 00000000000000000000. The Server host name is not given. The Boot file name is not given. The Magic cookie is DHCP. The Option (53) DHCP Message Type (Discover) section shows the length as 1 and the DHCP Discover (1) message type. The Option (61) Client identifier section shows the length as 7 and the hardware type as Ethernet (0x01). The Client MAC address is Grandstream_01:fc:42 (00:0b:82:01:fc:42). The Option (50) Requested IP Address (0.0.0.0) section shows the length as 4 and the requested IP address as 0.0.0.0. The Option (55) Parameter Request List section shows the length as 4 and the parameter request list items: (1) Subnet Mask, (2) Router, (3) Domain Name Server, and (4) Network Time Protocol Servers. The Option (255) End section shows the option end as 255 and the padding as 0000000000000000.

Structure:

- **Transaction ID:** 0x00003d1d (used to match requests and responses).
- **Your (client) IP address:** 192.168.0.10 (the IP address offered by the server).
- **Next Server IP address:** Not explicitly stated in the Discover, likely 0.0.0.0 or potentially an address like 192.168.0.4 if a specific boot server is involved.
- **Client MAC address:** 00:0b:82:01:fc:42 (the physical address of the client).
- **Options:** Include various parameters such as the Subnet Mask, Router (likely 192.168.0.1 based on the source IP of the Offer), Domain Name Server (likely an IP address on the network), and the IP Lease Time.

Let's analyze each option found in the first packet (DHCP Discover):

1. Option 53: DHCP Message Type

- **RFC 2132 Section 9.6**
- **Length:** 1

- **Content:** The message type is **Discover** (value 1).
- **Purpose:** This option specifies the type of DHCP message. In this case, it is a **Discover** message, which the client sends to find available DHCP servers on the network.

2. Option 61: Client Identifier

- **RFC 2132 Section 9.14**
- **Length:** 7
- **Content:**
- **Hardware Type:** Ethernet (1)
- **Client MAC Address:** 00:00:52:01:7c:42
- **Purpose:** This option is used to identify the client. According to RFC 2132, the Client Identifier can be the MAC address of the device or another unique identifier. In this case, the MAC address is used.

3. Option 50: Requested IP Address

- **RFC 2132 Section 9.1**
- **Length:** 4
- **Content:** 0.0.0.0
- **Purpose:** The client can request a specific IP address. According to RFC 2132, if the client does not request a specific address (as here, where the value is 0.0.0.0), the DHCP server will assign an available IP address from its pool

4. Option 55: Parameter Request List

- **RFC 2132 Section 9.8**
- **Length:** 4
- **Content:** The client requests the following parameters:
- **Subnet Mask (1):** The subnet mask (RFC 2132 Section 3.3).
- **Router (3):** The default gateway (RFC 2132 Section 3.5).
- **Domain Name Server (6):** The DNS server addresses (RFC 2132 Section 3.6).
- **Network Time Protocol Server (42):** The NTP server addresses (RFC 2132 Section 8.3).
- **Purpose:** This option specifies which parameters the client needs from the DHCP server. Each number in the list corresponds to a specific option as defined in RFC 2132.

5. Option 255: End

- **RFC 2132 Section 9.10**
- **Length:** 0
- **Content:** No content.
- **Purpose:** This option marks the end of the DHCP options. All DHCP options must end with this option.

6. Padding

- **RFC 2132 Section 9.11**
- **Content:** 00000000000000
- **Purpose:** Padding is used to fill the packet to meet protocol requirements. It does not contain useful information and is ignored during packet processing.

Packet 2:

[illegible]

Structure:

- **Transaction ID:** 0x080842d (used to match requests and responses).
- **Your (client) IP address:** 192.168.0.10 (the IP address offered by the server).
- **Next Server IP address:** 192.168.0.4 (the address of the next server, if applicable).
- **Client MAC address:** 00:00:82:01:fc:42 (the physical address of the client).
- **Options:** Include various parameters such as the subnet mask, DNS addresses, and the IP lease time.

Let's analyze each option found in the second packet (DHCP Offer):

1. Option 53: DHCP Message Type

- **RFC 2132 Section 9.6**
- **Length:** 1
- **Content:** The message type is **Offer** (value 2).
- **Purpose:** This option identifies the type of DHCP message. In your case, the value is 2, which corresponds to a **DHCP Offer**.

2. Option 1: Subnet Mask

- **RFC 2132 Section 3.3**
- **Length:** 4
- **Purpose:** This option specifies the subnet mask for the client's network. In your packet, the subnet mask is 255.255.255.0.

3. Option 58: Renewal (T1) Time Value

- **RFC 2132 Section 9.11**
- **Length:** 4
- **Purpose:** This option specifies the time interval (in seconds) after which the client should attempt to renew its IP address lease. In your packet, the value is 3000 seconds (30 minutes).

4. Option 59: Rebinding (T2) Time Value

- **RFC 2132 Section 9.12**
- **Length:** 4
- **Purpose:** This option specifies the time interval (in seconds) after which the client should attempt to rebind with another DHCP server if the original server is unavailable. In your packet, the value is 3150 seconds (30 minutes and 30 seconds).

5. Option 51: IP Address Lease Time

- **RFC 2132 Section 9.2**
- **Length:** 4
- **Purpose:** This option specifies the total duration (in seconds) for which the IP address is leased to the client. In your packet, the lease time is 3600 seconds (1 hour).

6. Option 54: DHCP Server Identifier

- **RFC 2132 Section 9.7**
- **Length:** 4
- **Purpose:** This option specifies the IP address of the DHCP server that sent the offer. In your packet, the server identifier is 192.168.0.1.
-

7. Option 255: End

- **RFC 2132 Section 9.10**
- **Purpose:** This option marks the end of the DHCP options list. It has no value or length.

Packet 3:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x3d1d
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x3d1d
3	0.070031	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x3d1e
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x3d1e

Dynamic Host Configuration Protocol (Request)

Message type: Boot Request (1)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x00003d1e
Seconds elapsed: 0

Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 0.0.0.0
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: GrandstreamM_01:fc:42 (00:0b:82:01:fc:42)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP

Option: (53) DHCP Message Type (Request)

Length: 1

DHCP: Request (3)

Option: (61) Client identifier

Length: 7

Hardware type: Ethernet (0x01)

Client MAC address: GrandstreamM_01:fc:42 (00:0b:82:01:fc:42)

Option: (50) Requested IP Address (192.168.0.10)

Length: 4

Requested IP Address: 192.168.0.10

Option: (54) DHCP Server Identifier (192.168.0.1)

Length: 4

DHCP Server Identifier: 192.168.0.1

Option: (55) Parameter Request List

Length: 4

Parameter Request List Item: (1) Subnet Mask

Parameter Request List Item: (3) Router

Parameter Request List Item: (6) Domain Name Server

Parameter Request List Item: (42) Network Time Protocol Servers

Option: (255) End

Option End: 255

Padding: 00

0000 ff ff ff ff ff ff 00 0b 82 01 fc 42 08 00 45 00B.E...
0010 01 2c a8 37 00 00 fa 11 17 8a 00 00 00 ff ff7.....
0020 ff ff 00 44 00 43 01 18 9f bd 01 01 06 00 00D.C.....
0030 3d 1e 00 00 00 00 00 00 00 00 00 00 00 00B.....
0040 00 00 00 00 00 00 0b 82 01 fc 42 00 00 00 00B.....
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0110 00 00 00 00 00 00 63 82 53 63 35 01 03 3d 07 01c.Sc5...
0120 00 0b 82 01 fc 42 32 04 c0 a8 00 0a 36 04 c0 a8B2....6...
0130 00 01 37 04 01 03 06 2a ff 007.....

Structure:

- **Transaction ID:** 0x3d1e (used to match requests and responses).
- **Your (client) IP address:** 192.168.0.10 (the IP address the client is requesting).
- **Next Server IP address:** 0.0.0.0 (no next server specified).
- **Client MAC address:** 00:0B:82:01:FC:42 (the physical address of the client).
- **Options:** Include various parameters such as the subnet mask, router (default gateway), DNS addresses, and network time protocol (NTP) servers.

Let's analyze each option found in the third packet (DHCP Request):

1. Option 53: DHCP Message Type

- **RFC 2132 Section 9.6**
- **Length:** 1
- **Value:** Request (3)
- **Purpose:** This option specifies the type of DHCP message. In this case, it is a DHCP Request, meaning the client is requesting to use the IP address offered by the DHCP server.

2. Option 61: Client Identifier

- **RFC 2132 Section 9.14**
- **Length:** 7
- **Value:**

Hardware Type: Ethernet (1)

MAC Address: 00:0B:82:01:FC:42
- **Purpose:** Used to uniquely identify the client. According to RFC 2132, the Client Identifier can be a MAC address or another unique identifier. Here, the client's MAC address is used.

3. Option 50: Requested IP Address

- **RFC 2132 Section 9.1**

- **Length:** 4
- **Value:** 192.168.0.10
- **Purpose:** The client requests a specific IP address. In this case, it asks for 192.168.0.10.

4. Option 54: DHCP Server Identifier

- **RFC 2132 Section 9.7**
- **Length:** 4
- **Value:** 192.168.0.1
- **Purpose:** Specifies the IP address of the DHCP server that offered the lease. Here, it is 192.168.0.1.

5. Option 55: Parameter Request List

- **RFC 2132 Section 9.8**
- **Length:** 4
- **Values Requested:**
 - Subnet Mask (1): Requests the subnet mask (RFC 2132 Section 3.3).
 - Router (3): Requests the default gateway (RFC 2132 Section 3.5).
 - Domain Name Server (6): Requests DNS server addresses (RFC 2132 Section 3.6).
 - Network Time Protocol Server (42): Requests NTP server addresses (RFC 2132 Section 8.3).
- **Purpose:** The client specifies which additional configuration parameters it needs from the DHCP server.

6. Option 255: End

- **RFC 2132 Section 9.10**
- **Length:** 0
- **Value:** None
- **Purpose:** Marks the end of the DHCP options. Every DHCP message must include this option at the end.

7. Padding

- **RFC 2132 Section 9.11**
- **Value:** Zeros for alignment
- **Purpose:** Used to ensure the packet meets protocol length requirements. It does not contain meaningful data.

Packet 4:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000090	0.0.0.0	255.255.255.255	DHCP	314	DHCP Discover - Transaction ID 0x3dd1
2	0.000295	192.168.0.1	192.168.0.10	DHCP	342	DHCP Offer - Transaction ID 0x3dd1
3	0.070831	0.0.0.0	255.255.255.255	DHCP	314	DHCP Request - Transaction ID 0x3d1e
4	0.070345	192.168.0.1	192.168.0.10	DHCP	342	DHCP ACK - Transaction ID 0x3d1e

User Datagram Protocol, Src Port: 67, Dst Port: 68

- Dynamic Host Configuration Protocol (ACK)
 - Message type: Boot Reply (2)
 - Hardware type: Ethernet (0x01)
 - Hardware address length: 6
 - Hops: 0
 - Transaction ID: 0x00003d1e
 - Seconds elapsed: 0
 - Bootp flags: 0x0000 (Unicast)
 - Client IP address: 0.0.0.0
 - Your (client) IP address: 192.168.0.10
 - Next server IP address: 0.0.0.0
 - Relay agent IP address: 0.0.0.0
 - Client MAC address: Grandstream_01:fc:42 (00:0b:82:01:fc:42)
 - Client hardware address padding: 00000000000000000000
 - Server host name not given
 - Boot file name not given
 - Magic cookie: DHCP
 - > Option: (53) DHCP Message Type (ACK)
 - Length: 1
 - DHCP: ACK (5)
 - > Option: (58) Renewal Time Value
 - Length: 4
 - Renewal Time Value: 30 minutes (1800)
 - > Option: (59) Rebinding Time Value
 - Length: 4
 - Rebinding Time Value: 52 minutes, 30 seconds (3150)
 - > Option: (61) IP Address Lease Time
 - Length: 4
 - IP Address Lease Time: 1 hour (3600)
 - > Option: (54) DHCP Server Identifier (192.168.0.1)
 - Length: 4
 - DHCP Server Identifier: 192.168.0.1
 - > Option: (1) Subnet Mask (255.255.255.0)
 - Length: 4
 - Subnet Mask: 255.255.255.0
 - > Option: (255) End
 - Option End: 255
 - Padding: 00

```

0000  00 0b 82 01 fc 42 00 00 74 ad f1 9b 08 00 45 00 .....B...t....E-
0010  01 48 04 46 00 00 80 11 00 00 c0 a8 00 91 c0 a8 ...H.F.....
0020  00 8a 00 43 00 44 01 34 df db 02 01 06 00 00 00 ...C.D.4.....
0030  3d 1e 00 00 00 00 00 00 00 00 c0 a8 00 0a 00 00 ==...B.....
0040  00 00 00 00 00 00 00 0b 82 01 fc 42 00 00 00 00 .....B.....
0050  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0090  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00a0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00b0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00c0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00d0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00e0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00f0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0100  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0110  00 00 00 00 00 00 63 82 53 63 35 01 05 3a 04 00 .....C.ScS.....
0120  00 07 08 3b 04 00 00 0c 4e 33 04 00 00 0e 10 36 ...;....N3.....6
0130  04 c0 a8 00 01 01 04 ff ff ff 00 ff 00 00 00 00 00 ..c.a..01.01.04.ffff...ff...00...00...00...
0140  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0150  00 00 00 00 00 .....
          
```

Bootp flags (dhcps), 2 byte(s)
Packets: 4
Profile: Defa

Structure:

- **Transaction ID:** 0x3d1e (used to match requests and responses). This matches the Wireshark capture "Transaction ID: 0x00003d1e".
- **Your (client) IP address:** 192.168.0.10 (the IP address being assigned to the client). This matches the Wireshark capture "Your (client) IP address: 192.168.0.10".
- **Next Server IP address:** 0.0.0.0 (no next server specified). This matches the Wireshark capture "Next server IP address: 0.0.0.0".
- **Client MAC address:** 00:0B:82:01:FC:42 (the physical address of the client). This matches the Wireshark capture "Client MAC address: GrandstreamN_01:fc:42 (00:0b:82:01:fc:42)".
- **Options:** Include various parameters configured by the DHCP server for the client.

Let's analyze each option found in the fourth packet (DHCP ACK):

1. Option 53: DHCP Message Type

- **RFC 2132 Section 9.6**
- **Length:** 1
- **Value:** ACK (5)
- **Purpose:** This option specifies the type of DHCP message. In this case, it is a DHCP ACK, confirming the client's request and providing configuration parameters. (Corrected from the initial description which incorrectly stated "Request (3)"). The Wireshark capture clearly shows "DHCP: ACK (5)".

2. Option 58: Renewal Time Value

- **RFC 2132 Section 9.13**
- **Length:** 4
- **Value:** 1800 (seconds), which is 30 minutes.
- **Purpose:** Specifies the time interval after which the client should contact the DHCP server to renew its lease. The Wireshark capture shows "Renewal Time Value: 30 minutes (1800)".

3. Option 59: Rebinding Time Value

- **RFC 2132 Section 9.13**
- **Length:** 4
- **Value:** 3150 (seconds), which is 52 minutes and 30 seconds.
- **Purpose:** Specifies the time interval after which the client should try to rebind to any DHCP server if it hasn't renewed its lease with the original server. The Wireshark capture shows "Rebinding Time Value: 52 minutes, 30 seconds (3150)".

4. Option 51: IP Address Lease Time

- **RFC 2132 Section 9.2**
- **Length:** 4
- **Value:** 3600 (seconds), which is 1 hour.
- **Purpose:** Specifies the duration for which the IP address is leased to the client. After this time, the lease expires. The Wireshark capture shows "IP Address Lease Time: 1 hour (3600)".

5. Option 54: DHCP Server Identifier

- **RFC 2132 Section 9.7**
- **Length:** 4
- **Value:** 192.168.0.1
- **Purpose:** Specifies the IP address of the DHCP server sending this ACK. In this case, it is 192.168.0.1. This matches the Wireshark capture "DHCP Server Identifier: 192.168.0.1".

6. Option 1: Subnet Mask

- **RFC 2132 Section 3.3**
- **Length:** 4
- **Value:** 255.255.255.0
- **Purpose:** Provides the subnet mask to the client, defining the network portion of the IP address. The Wireshark capture shows "Subnet Mask: 255.255.255.0".

7. Option 255: End

- **RFC 2132 Section 9.10**
- **Length:** 0
- **Value:** None
- **Purpose:** Marks the end of the DHCP options field. This is required for proper parsing of DHCP messages. The Wireshark capture shows "Option: (255) End".

8. Padding

- **RFC 2132 Section 9.11**
- **Value:** Zeros (0x00) for alignment.
- **Purpose:** Used to pad the packet to meet minimum length requirements or for alignment. The Wireshark capture shows "Padding: 0000000000000000...".